

LingoLand: An AI-Assisted Immersive Game for Language Learning

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Figure 1: Conceptual renderings of LingoLand using generated assets.

ABSTRACT

Immersion with a foreign language is key to increased motivation, satisfaction, and learning success. However, this can be hindered by anxiety and lack of access to immersive settings. LingoLand will address this by immersing people in foreign language learning environments designed to mimic real scenarios. Using generative machine learning, LingoLand presents players with personalized missions where they can freely interact with game characters, all endlessly patient and supportive of new language learners. Players receive instant feedback through a natural, voice-based interaction. Through LingoLand, players gain an understanding of different cultures while building practical language skills in a fun way.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Language Learning, Generative Games, Large Language Models, Human-AI Interaction

ACM Reference Format:

Olivia Seow. 2023. LingoLand: An AI-Assisted Immersive Game for Language Learning. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3625117>

1 INTRODUCTION

Learning a foreign language is often considered desirable and deeply rewarding. Immersive environments for foreign language learning allow learners to actively engage in communication and problem-solving tasks [8]. Such interactions, whether in-person or computer-mediated, have consistently been found to have enhanced pedagogical benefits [16, 17]. However, opportunities for immersion (e.g., encounters with a native speaker or travel to a foreign country) are not easily accessible. Even when present, people do not truly immerse themselves due to communication apprehension and fear of social evaluation [14]. This anxiety lowers motivation and self-confidence, further hindering successful language acquisition [4]. Game environments may hold keys to re-creating immersive scenarios sans such barriers to motivation and engagement [19].

An AI-assisted immersive web-based game, LingoLand reduces physical and psychological impedances to immersion by allowing learners to interact with generative agents and game assets optimized for foreign language learning in context.

Additionally, individualized instruction and learning paths have been shown to improve language learning gains [6]. LingoLand presents custom game-play to each learner, enabling such beneficial learning paradigms at scale.

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UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

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ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3625117>

2 GAME DESIGN

Genuine immersion requires a dynamic and adaptive environment. Games like Noun Town and L2 Speak approximate immersion in hard-coded game environments [2, 3]. Recently, Park et al. demonstrated that large language models can be used to create immersive virtual environments that simulate interpersonal communication, albeit without a first-person perspective engagement [15]. Generative models can also generate custom imagery that boosts memory retention [13]. Building on recent work, LingoLand will employ foundation models to derive endlessly engaging game scenarios and personalized learning paths in a simulated game environment. Distinctively, LingoLand introduces a first-person perspective to improve immersion in language learning.

To advance in LingoLand, players complete custom missions involving interacting with in-game objects and characters in a foreign language (see Figure 1). The missions can also be tied to real-world contexts across learning levels; they can range from buying a strawberry at a market, to discussing a prominent Human-Computer Interaction paper at a conference venue. New dialogue and assets can be generated when a new scenario requires it.

Speech-to-text technology will allow players to practice conversationally. *Broken* language inputs (e.g., mixtures of languages, incorrect pronunciation) can be understood by all in-game characters, who consistently provide nurturing guidance and correction, fostering an encouraging learning atmosphere.

3 TECHNOLOGY

LingoLand will use Anthropic’s Claude 2 to formulate sensible and contextually appropriate missions and conversational responses that support and stimulate the language learning journey. When the game begins, the user’s interests and language ability will be evaluated and integrated into a Claude 2 prompt to produce suitable missions.

In-game characters will each be crafted to be supportive of the player’s learning journey, while having a rich backstory that makes the game more dynamic and gratifying. Interaction between in-game characters enables natural game-play but does not need to happen frequently, helping to minimize API calls. A journal structure and event log will efficiently store memories in a vector database [1, 15].

OpenAI’s Whisper, an automatic speech recognition system, will process multi-lingual voice inputs. Since Claude 2 is cross-lingual, mixed language input can be directly parsed. For generating audio outputs, we intend to use MDN’s Web Speech, potentially employing other generated environmental sounds or natural voice generators for a more authentic feel.

Creating interactive visual elements is crucial for engaging game-play and memory retention. Hence, we plan to use a diffusion model with low-rank adaptation [10] to craft game assets, resembling those conceptualized in Figure 1. Players can also create custom flashcards using a doodle-based ControlNet [20]. This aids memory recall and helps track players’ progress across individual learning styles and preferences.

The front-end will use React.js and TypeScript for creating an interactive user interface that is broadly accessible to many users via a web browser.

Collectively, these modern tools enable us to make LingoLand an engaging, effective, and enjoyable language learning experience.

4 DEVELOPMENT PROCESS

To improve design outcomes, LingoLand will undergo iterative development through a parallel prototyping process [7]. This approach will be complemented by the adoption of human-centered design practices, including in-depth needs investigation and regular user feedback sessions [12]. We will also refer to proposed best practices on human-AI interaction for further guidance [5].

From the early stages of development and as abiding by Institutional Review Board protocols, language learners from diverse backgrounds will be invited to test the game’s design and functionality. Their input will guide the refining of game mechanisms, missions, personalized learning capabilities, and an appropriate framework to measure LingoLand’s success. We will also consult with human game designers to better understand the role and impact of AI-assisted game design.

We expect that LingoLand will evolve as we receive user insights that provide a deeper understanding of how different people engage with the game. The development process will allow us to create a tool sensitive to diverse needs, as elaborated in Section 5. Throughout development and deployment, we will continuously collect and integrate user feedback to improve and refine LingoLand.

5 ETHICS AND BROADER IMPACT

LingoLand aims to offer an experiment in achieving enhanced language learning through massive customization via human-AI *collaboration*. It is not a proposal to eliminate designers, teachers, or human interactions altogether. LingoLand will not attempt to cover all aspects of language learning. We hypothesize that platforms like LingoLand are even more effective when used with actual human instruction and interaction.

While large language models have been shown to excel at many multi-lingual tasks [18], this does not apply equally well across all use cases [21]. Moreover, we also acknowledge that it would be nearly impossible to recreate the richness of reality and nuances of diverse cultures in a game environment. We will remain vigilant to ensure the game is sensitive to various cultures and emphasizes inclusivity across a broad range of users.

As Kukulska-Hulme et al. describe, it is crucial for AI-assisted language learning tools to accept a diverse range of voice inputs and to avoid prejudicial or false generations [11]. While LingoLand is proposed in the spirit of play, we strive to foster a respectful environment for diverse players. We will take steps to mitigate unnatural, discriminatory, disrespectful, insensitive, and inappropriate interactions.

6 CONCLUSION AND NEXT STEPS

LingoLand promises new avenues of research towards effective and engaging language learning methodologies. If successful, we envision extending LingoLand from a web-based game to one in virtual reality, a modality of immersion that has been shown to increase “attention, motivation, and enjoyment” in foreign language learning [9]. We aim to contribute to the development of a truly immersive language learning tool.

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